HW I

EE 588: Optimization for the information and data sciences

University of Southern California

Assigned on: August 26, 2026 Due date: beginning of class on September 9, 2026

The following exercises from the text: 2.2., 2.5, 2.7, 2.10, 2.28, 3.2, 3.3 and

convex hulls

(a) A vector $\boldsymbol{x} \in \mathbb{R}^n$ is 1-sparse if it has only one non-zero entry. What is the convex hull of all 1-sparse vectors in $\mathbb{R}^n$ with unit Euclidean norm? Justify your answer.

(b) What is the convex hull of all orthogonal matrices? Justify your answer.